Patrick Engelbert

patrickengelbert.com

ROBOTICS • AUTOMATION • INDUSTRIAL CONTROLS • SOFTWARE ENGINEERING

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PROFESSIONAL PROFILE

Automation and industrial controls professional working directly with engineering teams at MP Materials on programming, design, maintenance, and troubleshooting. Former full-stack software engineer with production development experience at Citizens Bank. Completing an associate degree in Robotics & Industrial Controls Technology in December 2026.

TECHNICAL STRENGTHS

PLCs & controls: Rockwell Studio 5000, ladder logic, relay circuits, industrial I/O, timers, interlocks, sequencing, troubleshooting

Robotics & integration: FANUC 200i, Universal Robots / PolyScope, teach pendants, position registers, motion offsets, robotic pick-and-place, sensors

Electrical & design: AC/DC circuits, motor controls, VFDs, HMIs, electromechanical devices, schematic reading, AutoCAD, system integration

Software & teamwork: Python, TypeScript, JavaScript, Angular, React, Node.js, SQL, Git, Agile, technical documentation, cross-functional collaboration

PROFESSIONAL EXPERIENCE

Facilities & Maintenance Intern (Engineering Projects) | MP Materials **Aug 2026 – Present**

Fort Worth, TX

- Work directly with engineering teams on a daily basis, contributing to equipment design, automation programming, system implementation, and maintenance activities.
- Program and test industrial robotic systems, developing logic for sensing, positioning, automated handling, and repeatable motion sequences.
- Participate in engineering planning and design meetings; contribute ideas on system architecture, equipment functionality, and practical implementation.
- Support troubleshooting and maintenance of manufacturing equipment and automation systems, applying hands-on diagnostic and problem-solving skills.

Full-Stack Software Engineer | Citizens Bank **Mar 2022 – Nov 2024**

Johnston, RI

- Developed and maintained production web applications and customer-facing components using Angular, TypeScript, and Node.js.
- Built proprietary functionality for secure collection of customer data and contributed core components to citizensbank.com.
- Collaborated on an Agile engineering team to design, review, test, document, and deliver software under production requirements.

EDUCATION & CURRENT CAPSTONE

Associate of Applied Science, Robotics & Industrial Controls Technology **Dec 2026 (expected)**

Texas State Technical College | Waco, TX

- Final-semester integrated automation capstone (in progress): combining a PLC, HMI, VFD, sensors, and industrial robot into one coordinated control system.

SELECTED AUTOMATION PROJECTS

FANUC 200i - Nine-Position Pick-and-Place | Robotics / TSTC

- Programmed one reference position and computed the remaining eight positions with offsets and position registers, reducing manual teaching.
- Verified repeatability by running the full nine-position sequence overnight in automatic mode without faults or collisions.

Studio 5000 PLC Arcade | Rockwell / Ladder Logic

- Programmed an eight-light reflex game with operator pushbuttons, timer-driven sequencing, and six selectable difficulty modes.
- Implemented randomized timing in turbo modes to demonstrate state management, dynamic sequencing, and reliable I/O logic.

Sensor-Controlled Conveyor | Industrial Circuits / Documentation

- Designed a start/stop latch and inductive-sensor interlock to pause conveyor motion on detection and resume automatically when clear.
- Prepared circuit schematics, parts list, system purpose, and sequence-of-operation documentation for maintenance and troubleshooting.

Universal Robots - Automated Workpiece Handling | Industrial Robotics

- Programmed robotic motion and feedback-driven handling sequences using tool I/O, position sensing, and vacuum pickup verification.

ADDITIONAL WORK EXPERIENCE

Pharmacy Technician | Cosentino's Price Chopper **Jun – Sep 2021**

Ottawa, KS

- Coordinated with pharmacy personnel to dispense medications accurately while safeguarding sensitive customer information.

After-School Teacher | RFTS / USD 290 **Aug 2019 – Apr 2020**

Ottawa, KS

- Planned and delivered instruction; collaborated with colleagues on lessons, projects, and clear learning objectives.

Sales Associate | Walmart Supercenter **Apr 2015 – Jun 2019**

Ottawa / Salina, KS

- Maintained nearly four years of retail employment, supporting customer needs, product operations, inventory, and team coordination.

Bank Teller | Kansas State Bank **Apr 2014 – Mar 2015**

Ottawa, KS

- Handled financial transactions accurately and built dependable relationships with individual and business customers.

ADDITIONAL EDUCATION & CREDENTIALS

Software Engineering Program | Thinkful | 2021–2022

Software Engineering Certificates | Three completed certificates

Associate of Science & Associate of Arts, Philosophy | Neosho County Community College | 2020

Undergraduate Studies, Philosophy / Theological Studies | Newman University | 2020–2021